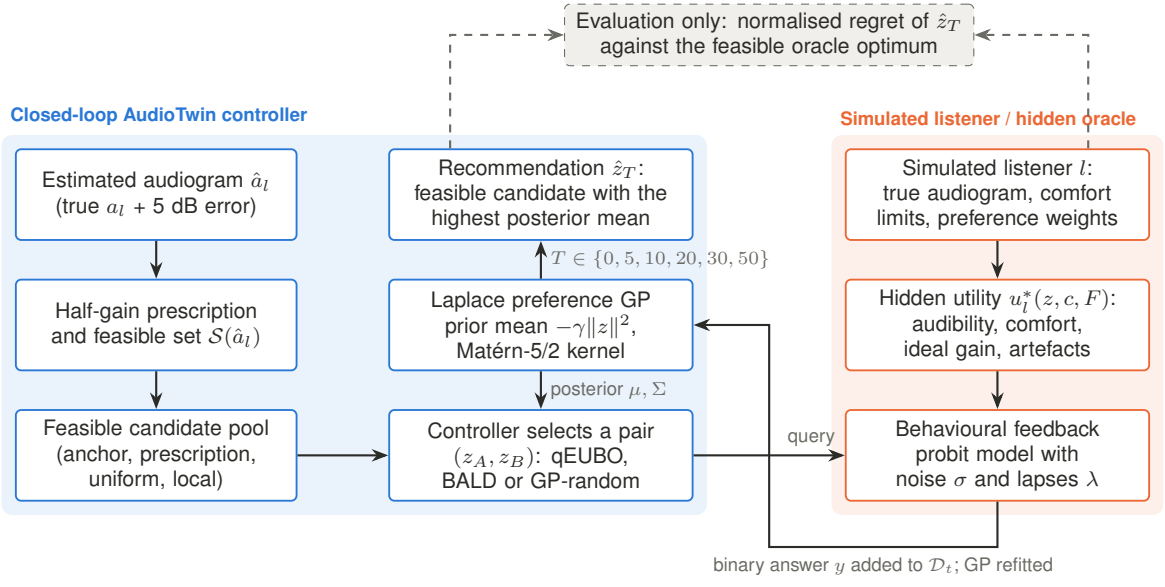

Evaluation only: normalised regret of $\hat{z}_T$ against the feasible oracle optimum
Closed-loop AudioTwin controller
Simulated listener / hidden oracle
Estimated audiogram $\hat{a}_l$ (true a_l + 5 dB error)
Half-gain prescription and feasible set $\mathcal{S}(\hat{a}_l)$
Feasible candidate pool (anchor, prescription, uniform, local)
Recommendation $\hat{z}_T$: feasible candidate with the highest posterior mean
$T \in \{0, 5, 10, 20, 30, 50\}$
Laplace preference GP prior mean $-\gamma\|z\|^2$, Matérn-5/2 kernel
posterior μ, Σ
Controller selects a pair (z_A, z_B): qEUBO, BALD or GP-random
query
Simulated listener l: true audiogram, comfort limits, preference weights
Hidden utility $u_l^*(z, c, F)$: audibility, comfort, ideal gain, artefacts
Behavioural feedback probit model with noise σ and lapses λ
binary answer y added to $\mathcal{D}_t$; GP refitted